

Category	Name	Licence	Description	Latest features
SCADA	IndigoSCADA	Free	Small-footprint SCADA system entirely developed in C and C++ with multiple OS support and multiple front end protocol drivers	
SCADA	S.E.E.R.2.1.5.4	Free		- Moved to phpsinfo version 3, supporting modern Windows 7 or Server 2008 and up - Ability to give touch panel instances a friendly name, rather than only go by their machine name (panel name) for user display - Added ON_OFF_IND_TOGGLE_INVERSE and ON_OFF_IND_INVERSE to the touchpanel
Schematic editor, mixed signal simulator and power consumption estimator	SLED 2.4.0, SMASH 6.4.0 and SCROOGE 3.4.0	Licensed	Hierarchical schematic editor SLED, mixed-signal simulator SMASH and extension SCROOGE for power consumption analysis	
Signal processing	GNU Radio 3.7.5	Free	C language based software development toolkit consisting of signal-processing blocks for creating software-defined radios and other signal-processing systems	- Added snap-to-grid, hiding of port names (optional) and saving of console output to GNU Radio Companion (creates flowgraphs), and newly-created or saved GRC files will have version information embedded into these - Two separate graphics systems for producing display output, based on wxWidgets and QT windowing system, have reached feature parity with each other - New digital television component that incorporates receiver code
Simulator	DoCircuits	Dual	Online virtual electronics lab with which you can prototype and test designs	Available on Windows 8.1 devices (tablets, ultrabooks and the like)
Simulator	Archimedes 2.0.1	Free	For design, simulation and behaviour prediction of sub-micron and mesoscopic semiconductor devices, based on the ensemble Monte Carlo method	Fixes a small problem related to Automake vulnerabilities
Simulator	Ltspice IV (formerly SwitcherCAD)	Free	SPICE tool to virtually model, simulate and analyse electronic circuits, particularly switching regulators	- Supports Mac OS X 10.7+ platforms - Multi-threaded solvers that utilise multi-core processors - New SPARSE matrix solvers that deploy self-authoring code

want to add Internet connectivity to it. In the first stage, you can place the collected data on the cloud, and in the second stage, you would need an application running locally on the device to control the temperature and set up a Web interface to it. Then, you would need to set up locations for the device, a smartphone, for example, to control the system. The designer needs infrastructure to set up a server or a portal or a platform, where the mobile phone is connecting to and syncing with the cloud.

More tools in the sky

Another interesting trend in the software tools arena is the growing number and increasing popularity of Web based development tools for embedded systems.

Jason Guo, senior technical staff

India in the picture

The software development tools sector in Indian electronics industry is growing fast, mainly because of the good number of established independent design houses and the present start-up wave here. It is interesting to see that there are a few software development tools coming up from India too.

Experiment in your virtual lab. Sakshat Virtual Labs, an initiative by Ministry of Human Resource and Development (MHRD), is one of the many interesting projects that came up from India. It is a set of free virtual lab software developed by 12 different educational institutes including the IITs and is undergoing continuous developments. These let you model physical phenomena with equations and simulate these to predict and understand the approximate result of the experiment in real. These also let you remotely trigger an experiment in a real lab and get results through a computer interface.

Going online. Bengaluru based Sparsha Learning Technologies Pvt Ltd has come up with DoCircuits, a website that provides users with a virtual electronics lab that allows prototyping and testing of electronics circuits, storing these on the cloud and accessing these from anywhere securely.

member of Fairchild Semiconductor, recently said in an interaction with EYF that some advanced electronic design and simulation tools are now available online to enable system-level design of switching mode power

supply (SMPS).

Cloud based development tools reduce the hassle of setting up software, resolve space issues in the user's computer and avoid frequent upgrades of the tool. These also al-